

TA(s) Consulted: [Name(s) of TA(s) you spoke to]

Instrument Cataloguer

1 Team Members

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2 Task Overview

We are creating a program to catalogue various percussive instruments based on their waveform. Many musicians find it very difficult to catalogue instruments because many instruments do not have physical identifiers on them.

This task is challenging because the raw waveform must be converted with a Fourier transform into time–frequency features such as a spectrogram.

3 Task Definition

The data we are working with is an audio .mp3 file. This is a single-label, multi-class classification task with 20 classes, one per instrument in the dataset: flute, oboe, cor anglais (English horn), clarinet, bass clarinet, bassoon, contrabassoon, saxophone, French horn, trombone, trumpet, tuba, banjo, cello, double bass, guitar, mandolin, viola, violin. Each clip is one note by one instrument. We have multiple features: fundamental frequencies, timbre, resonance, harmonics, noisiness/dynamics.

4 Problem Description and Impact

Suppose a recording of a composition uses a certain wind/string instrument, and we want to find the same instrument again. Being able to say what instrument a sound came from based on how it sounds makes it easier to search.

The challenges are: (1) audio analysis requires knowledge of signal processing; we will be using Fourier transforms; (2) the same instrument sounds different across pitches, dynamics and articulation, so the model must not just learn “pitch” or “loudness” as a shortcut; (3) many instruments in the same family share very similar timbre; (4) because the recordings are really clean, this may not transfer to real-world recordings.

5 Data Source(s) and Collection Plan

[Describe where your data comes from. Include:

- Links to the data source(s):
- <https://philharmonia.co.uk/resources/sound-samples/>
- Metadata: each file's name has the instrument, pitch, note duration, dynamics and articulation
- Data is already labeled
- If not collected by you: details on how it was originally collected/annotated/preprocessed
- We will be using a subset of the full database (brass, strings, woodwinds). This is because percussion instruments do not have that many recordings.

6 Dataset Size and Example Data Points

The collection of instruments has well over 1000 data points (one note per file); once we parse the data, we will report the exact count.

Example Data Points

1. **Input:** [flute_A4_1_forte_normal.mp3 (a 1-second flute note A4, forte)] **Label:** [flute]
2. **Input:** [clarinet_C4_1_piano_normal.mp3 (clarinet note, C4, piano)] **Label:** [clarinet]
3. **Input:** [bassoon_G2_1_fortissimo_normal.mp3 (low bassoon note, G2)] **Label:** [bassoon]

7 Proposed Solution

- Each clip is resampled, trimmed, normalized, then converted to a spectrogram. The target label is one of the 19 instruments.
- What model(s) will you try (no simple linear regression)?
- Instrument recognition is well studied. Our project applies these ideas to a focused instrument subset.
- How will you evaluate whether your model is good (metrics, validation strategy)?
- librosa, torchaudio/PyTorch, scikit-learn, pandas, NumPy, and matplotlib

Sources of Inspiration

1. J. Engel et al., “Neural Audio Synthesis of Musical Notes with WaveNet Autoencoders” (the NSynth dataset), ICML 2017.

2. J. J. Bosch, J. Janer, F. Fuhrmann, and P. Herrera, “A Comparison of Sound Segregation Techniques for Predominant Instrument Recognition in Musical Audio Signals” (the IRMAS dataset), ISMIR 2012.
3. Y. Han, J. Kim, and K. Lee, “Deep Convolutional Neural Networks for Predominant Instrument Recognition in Polyphonic Music,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2017.
4. Q. Kong et al., “PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2020.

8 TA Consultation Summary

[Summarize what you discussed with your TA, the advice/feedback they provided, and how it influenced or changed your project plans.]